Math 564 – Friday Sept 5

Conjugate Gradient Methods (line search)

Consider the optimization problem

$$(*) \quad \min \; f(x) = \tfrac{1}{2} x^T A x - b x \quad \text{s.t. } x \in \mathbb{R}^n$$

quadratic
bounded below
unique min

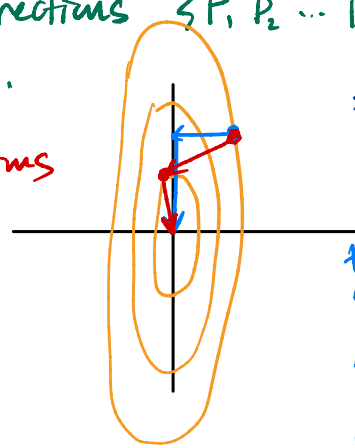
where A is symmetric pos. def. and $b \in \mathbb{R}^n$.

The unique solution is where $\nabla f = 0$

$$\nabla f(x) = Ax - b \overset{\text{set}}{=} 0 \quad \Rightarrow \text{ solution to } Ax = b.$$

This optimization problem is a fancy way to solve $Ax = b$.

Consider any set of conjugate directions $\{p_1, p_2 \dots p_n\}$ where $p_i^T A p_j = 0$ when $i \neq j$.



example:

$\frac{1}{2} x^T A x$

$A = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$

Using a set of conjugate directions we can solve $(*)$ in at most n iterations (with exact line searches).

If we knew the orientation of the level sets we could choose n appropriate directions and minimize in n iterations or fewer.

But!! conjugate directions also have this property.

exact line search

$$\left\{ \begin{aligned} x_{k+1} &= x_k + \alpha_k p_k \\ \alpha_k &= \underset{\beta}{\operatorname{argmin}} \; f(x_k + \beta p_k) \end{aligned} \right\}$$

$$f(x_k + \beta p_k) = \tfrac{1}{2} (x_k + \beta p_k)^T A (x_k + \beta p_k) - b^T (x_k + \beta p_k)$$

$$\frac{\partial f}{\partial \beta} = x_k^T A p_k + \beta p_k^T A p_k - b^T p_k \overset{\text{set}}{=} 0$$

$$\alpha_k = \frac{-p_k^T (A x_k - b)}{p_k^T A p_k} = \frac{-p_k^T r_k}{p_k^T A p_k} \leftarrow \text{step length}$$

A conjugate gradient algorithm for linear systems

1. Given: x_0
2. Set: $r_0 = Ax_0 - b$, $p_0 = -r_0$, $k=0$
3. Compute the next iterate

$$\alpha_k = \frac{-r_k^T p_k}{p_k^T A p_k} = \frac{\|r_k\|^2}{\|p_k\|_A^2}$$

$$x_{k+1} = x_k + \alpha_k p_k$$

from the relations $r_k^T p_j = 0$ $r_k^T r_j = 0$ $p_k^T A p_j = 0$

4. Updates

$$r_{k+1} = A x_{k+1} - b$$

$$\beta_{k+1} = \frac{r_{k+1}^T A p_k}{p_k^T A p_k} = \frac{\|r_{k+1}\|_A^2}{\|r_k\|_A^2}$$

$$p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$$

$$k = k+1$$

choose $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$

Apply $p_{k+1} A p_k = 0$

5. If $r_k = 0$ then stop otherwise go to step 3

This algorithm has some nice properties.

- converges to a solution with m iterations wher m is the number of unique nonzero eigenvalues of A.
- p_{k+1} only depends on p_k (not $p_1 p_2 \cdots p_k$).
- Low memory footprint, low computational cost

Nonlinear Conjugate Gradient Method

- residual r_k replaced by the gradient $\nabla f(x_k)$
- step length α_k determined by inexact line search

(Fletcher–Reeves)

1. Given: x_0, $\varepsilon > 0$
2. set $p_0 = -\nabla f(x_0)$, $k \leftarrow 0$
3. If $\|\nabla f(x_k)\| < \varepsilon$ then stop
4. Compute $\alpha_k \approx \underset{\beta}{\operatorname{argmin}}\, f(x_k + \beta p_k)$ ← perform line search with strong wolfe cond. with $0 < c_1 < c_2 < 1/2$. (See lemma 5.6)
5. Updates

$$x_{k+1} = x_k + \alpha_k p_k$$

$$\beta_{k+1} = \frac{\|\nabla f(x_{k+1})\|^2}{\|\nabla f(x_k)\|^2}$$

$$p_{k+1} = -\nabla f(x_{k+1}) + \beta_{k+1} p_k$$

$$k \leftarrow k+1$$

6. Go to step 3

Also require a restart every n iterations.

Polak–Ribière method

$$\beta_{k+1} = \frac{\nabla f(x_{k+1})^T \left(\nabla f(x_{k+1}) - \nabla f(x_k)\right)}{\nabla f(x_k)^T \nabla f(x_k)}$$

$$\beta_{k+1} = \max\{\beta_{k+1}, 0\}$$

provides a natural restart mechanism

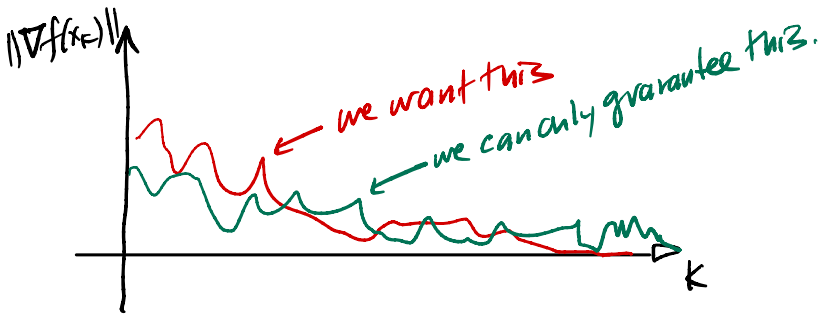
Assumption : $\mathcal{L} = \{ x \in \mathbb{R}^n \mid f(x) \le f(x_0) \}$ is bounded

Assumption : In $\mathcal{N} \supset \mathcal{L}$, f is Lip. cont. diff.

Theorem Suppose the assumptions and CG algorithm with S.W. cond, $0 < c_1 < c_2 < 1/2$. Then

$$\liminf_{k \to \infty} \|\nabla f(x_k)\| = 0$$

This convergence result is "ok". We would like to have the stronger result $\lim_{k \to \infty} \|\nabla f(x_k)\| = 0$.



There exists subsequence $\{w_k\}$ of $\{x_k\}$ such that $\lim_{k \to \infty} \|\nabla f(w_k)\| = 0$.